

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
Student Name:	ALLU THANMAI	Reg. No.:	192412616
Section / Batch:	ECE	Date:	09-08-2026

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

ANS:

Object Boundary Detection in Medical Images

(a) Interpretation and Importance of Boundary Detection

In medical image processing, **boundary detection** refers to identifying the edges or borders that separate different anatomical structures or tissues in an image. For example, in an MRI, CT scan, or ultrasound image, boundaries may exist between a tumor and surrounding healthy tissue, between organs, or between different types of tissue.

Boundary detection is important because it helps:

- **Segment anatomical structures** accurately.
- **Detect and locate tumors or lesions.**
- Determine the **size, shape, and position** of abnormal regions.
- Assist doctors in **diagnosis and treatment planning.**
- Support **computer-aided diagnosis** and automated medical image analysis.
- Provide clearer visualization of structures that may otherwise be difficult to distinguish.

Accurate boundaries are therefore essential for reliable interpretation of medical images.

(b) Key Challenges in Boundary Detection

Several factors make boundary detection in medical images difficult:

1. **Low contrast:** Different tissues can have very similar intensity values, making their boundaries weak.
2. **Noise:** Medical images often contain noise caused by the imaging equipment or acquisition process.
3. **Illumination/intensity variations:** Intensity may vary within the same tissue because of imaging conditions.
4. **Blurred boundaries:** Some tissue boundaries are not sharp and may gradually transition from one tissue type to another.
5. **Complex anatomical structures:** Organs and tissues may have irregular or complicated shapes.
6. **Artifacts:** Motion, imaging artifacts, and distortions can create false edges.
7. **Small or weak structures:** Important abnormalities may produce very subtle edges that are difficult to detect.

(c) Suitable Edge Detection Method: Canny Edge Detector

A suitable method for this application is the **Canny edge detection algorithm.**

The main steps of Canny edge detection are:

1. **Gaussian smoothing:**

The image is first smoothed using a Gaussian filter to reduce noise.

2. **Gradient calculation:**

The intensity gradient is calculated to identify areas where the image intensity changes rapidly.

3. **Non-maximum suppression:**

Thick edge regions are reduced to thin, well-localized edges.

4. **Double thresholding:**

Two thresholds are used to classify pixels as strong edges, weak edges, or non-edges.

5. **Edgetracking by hysteresis:**

Weak edges are retained when they are connected to strong edges, while isolated weak edges are discarded.

The output is an image containing the significant boundaries between tissues.

(d) Justification for Selecting Canny Edge Detection

Canny is particularly appropriate for medical images because it provides a good balance between **noise reduction, accurate localization, and detection of weak boundaries**.

- **Noise suppression:** Gaussian smoothing reduces the effect of image noise before detecting edges.
- **Good localization:** Non-maximum suppression produces thin and well-defined boundaries.
- **Reduced false edges:** Double thresholding and hysteresis help eliminate insignificant noise-related edges.
- **Detection of weak boundaries:** Weak tissue boundaries can be retained if they are connected to meaningful strong edges.
- **Adaptability:** The threshold values and Gaussian filter size can be adjusted depending on the type and quality of the medical image.

Compared with a simple Sobel or Prewitt operator, Canny generally provides cleaner and more continuous boundaries, which is valuable when identifying anatomical structures.

(e) Systematic Procedure

The overall boundary-detection process can be summarized as:

Medical Image → Noise Reduction → Gradient Calculation → Non-Maximum Suppression → Double Thresholding → Edge Tracking → Detected Tissue Boundaries

For example, if the objective is to identify the boundary of a tumor in an MRI image, the MRI image is first smoothed to reduce noise. Canny edge detection is then applied to identify intensity changes around the tumor. The resulting edge map can subsequently be used by a segmentation algorithm to isolate and measure the tumor.

Conclusion

Boundary detection is a fundamental step in medical image analysis because it enables the separation and identification of different tissues and abnormal structures. Although medical images contain challenges such as noise, low contrast, blurred boundaries, and artifacts, the **Canny edge detector** is a suitable choice because it combines noise reduction, accurate edge localization, and effective suppression of false edges. Therefore, it can provide reliable boundary information for subsequent medical image segmentation, diagnosis, and analysis.

Code of Conduct

/ ALLU THANMAI(192412616) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **thanmai A**

RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	